

AI-Assisted Automatic Incident Summarization from SIEM Logs

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Abstract— Today, cybersecurity operations centers (SOCs) require automated and interpretable systems to analyze large volumes of log data quickly, accurately, and in a centralized manner. In this study, a security automation system that enables automatic analysis of SIEM logs with artificial intelligence support is presented. In the proposed system, incident information entered by users is collected through Google Forms, transferred to the n8n workflow platform via Google Sheets, and processed automatically. Within the workflow, a Groq-based language model is used in the artificial intelligence analysis layer to evaluate log records and generate incident summaries, attack type classifications, risk scores, and recommended response actions. The generated analysis results are centrally stored in Airtable database tables including Incidents, Users, AI Analysis History, and Threat Intelligence. In addition, automatic email notifications regarding incident submission and analysis results are sent to users through Gmail integration. In the test scenarios conducted, the system successfully classified different cybersecurity events such as brute force attacks, malware activities, suspicious VPN logins, unauthorized access attempts, and DDoS attacks. Experimental results demonstrate that the developed system reduces manual analysis processes, accelerates incident management, and provides a practical solution for AI-supported security automation.

Keywords— SIEM, Cybersecurity, Artificial Intelligence, Log Analysis, n8n, Airtable, Incident Response, Security Automation.

I. INTRODUCTION

In today's digital world, cyberattacks targeting corporate network infrastructures, servers, user accounts, and critical systems are increasing every day. Modern attackers often perform long-term reconnaissance activities before directly accessing target systems, analyzing network structures and identifying vulnerable areas within systems. During this process, numerous security events such as failed login attempts, unusual connections, unauthorized access attempts, and malware activities may occur. Detecting these incidents at an early stage and evaluating them accurately is crucial for maintaining the security level of organizations.

One of the most important data sources in cybersecurity processes is system log records. Log data generated by network devices, firewall systems, authentication services, servers, and applications provide security teams with detailed visibility into events occurring within the system. Through these records, attack attempts, abnormal user behavior, and potential threats

can be identified. However, in large-scale systems, the excessive volume of log data makes manual analysis difficult for security teams.

Security Information and Event Management (SIEM) systems, which are widely used in organizations, collect log records from multiple sources in a centralized structure. These systems enable security events to be monitored from a single point. However, as the number of logs increases, manually reviewing every record becomes time-consuming for analysts and may delay the detection of critical incidents. Rapid evaluation is especially important for attacks such as brute force attempts, malware activities, suspicious user logins, and distributed denial-of-service attacks.[1]

Traditional log analysis processes create operational workload and may cause critical records to be overlooked within large volumes of data. In addition, checking records individually across different platforms complicates process management. Therefore, making log analysis faster, centralized, and automated has become an important requirement in the field of cybersecurity.

In this study, a security automation system that enables automatic analysis of SIEM logs with artificial intelligence support was developed. In the proposed system, security incidents entered by users are collected through Google Forms, transferred to the n8n workflow platform via Google Sheets, and processed automatically. Within the workflow, a Groq-based artificial intelligence model analyzes the log records and generates an incident summary, attack type classification, risk score, and recommended response actions. The obtained results are centrally stored in Airtable, and automatic notifications are sent to users through Gmail integration.

In this context, the main contributions of this study to cybersecurity and security automation can be summarized as follows:[2]

n8n-Based Automatic SIEM Log Processing and Workflow Management: Security incidents entered by users are collected through Google Forms and transferred to the n8n workflow structure via Google Sheets. The automation built within the workflow processes incident records automatically without manual intervention and manages each step in a structured sequence. This architecture enables faster evaluation of security incidents.

AI-Assisted Automatic Attack Analysis and Risk Assessment: SIEM log records are analyzed using a Groq-based artificial intelligence model to determine the type of incident and generate a risk score. The system not only produces a summary but also classifies different attack categories such as brute force, suspicious login, malware activity, unauthorized access, and DDoS, providing security teams with rapid preliminary assessments.[3]

Centralized Incident Tracking and Record Management with Airtable: The Airtable database includes Incidents, Users, AI Analysis History, and Threat Intelligence tables. This structure allows security incidents to be stored centrally, user-based records to be monitored, and previous analysis results to be reviewed in an organized manner.[4]

Automatic Notification and User Communication Integration: Through Gmail integration within the workflow, users are automatically informed when an incident is received and when analysis is completed. As a result, the system not only performs analysis but also provides end-to-end user communication as part of the incident management process.

As a result, this study combines AI-assisted SIEM log analysis, workflow automation, and centralized incident management within a single structure, providing a practical system that accelerates security operations and reduces manual workload.

II. SYSTEM ARCHITECTURE

The main objective of the developed security automation system is to automatically process SIEM log records entered by users without requiring manual intervention, transfer them to the artificial intelligence analysis layer, store the generated results in a centralized database, and notify users quickly. In this context, the system was designed with an integrated architecture in which data collection, automation, artificial intelligence analysis, database management, and user notification processes operate together.

Within the system, incident information collected through Google Forms is transferred to Google Sheets, processed by the n8n workflow platform, analyzed by a Groq-based artificial intelligence model, and stored in the Airtable database. At the end of the process, an automatic email notification is sent to the user through Gmail integration. This structure enables data flow to proceed systematically and allows all security incidents to be managed centrally.

The main layers that constitute the software and data flow architecture of the system are explained below.

A. Data Input Layer and Incident Collection Process

In the first layer of the system, users enter security incident information through Google Forms. This form collects data such as user name, email address, department, incident title, priority level, and SIEM log content in a standardized structure.

Since form data is collected through predefined fields, the data entry process is organized and incomplete or inconsistent entries are minimized. As soon as the user submits the form, the information is automatically transferred to the Google Sheets table.

At this stage, Google Sheets functions as the data source and serves as the starting point of the workflow system. In this way, the system can detect new records without requiring manual execution.[5]

B. n8n-Based Automation and Workflow Management

The main operational structure of the system was developed on the n8n workflow platform. Whenever a new record is added to the Google Sheets table, the Trigger node automatically starts the workflow process.

The nodes within the workflow operate in a specific sequence. First, an incident record is created, then user information is stored, and an initial notification email is sent. In the next step, the SIEM log record is transferred to the artificial intelligence analysis layer.

After the analysis is completed, the generated results are stored in the database and the existing incident record is updated. At the end of the process, a second email containing the analysis result is sent to the user.

This structure allows data flow to be managed from a single center and ensures that every incident progresses automatically through the same processing steps.

Fig. 1. n8n workflow-based system architecture and data flow

C. AI-Assisted Log Analysis Layer

One of the most important layers of the system is the artificial intelligence analysis component. SIEM log records received from the workflow are sent to the Groq-based language model.

The artificial intelligence model analyzes the log content, determines the incident type, generates a short incident summary, and produces a risk score. In addition, recommended response actions that can be applied by security teams are also generated by the system.

This structure reduces the need for manual log review and enables security incidents to be evaluated more quickly. Furthermore, the system helps identify different attack categories and allows incident priority levels to be understood more efficiently.[6]

D. Database Management and Record Association

The Airtable database used in the system was designed to centrally store incident records. The database includes the Incidents, Users, AI Analysis History, and Threat Intelligence tables.

While incident records are stored in the Incidents table, user information is stored in the Users table. Analysis outputs generated by artificial intelligence are saved in the AI Analysis History table, and threat-related data is stored in the Threat Intelligence table.

Relationships between tables are established through linked record fields. In this way, it becomes possible to centrally view which user created a specific incident and which AI analysis is associated with it.[7]

E. Notification and User Communication Layer

In the final layer of the system, Gmail integration is used. When the workflow begins, the user receives an automatic notification email confirming that the incident record has been successfully received.

After the artificial intelligence analysis is completed, a second email containing the incident summary, attack type, risk score, and recommended response actions is sent to the user.

Through this structure, users are informed automatically throughout the entire process without requiring manual follow-up. As a result, incident management becomes more organized and centralized from both the technical and user perspectives.[8]

III. DETECTION METHODS AND ALGORITHMS

The developed AI-assisted SIEM log analysis system uses a multi-layered analysis approach to accurately classify security incidents, determine risk levels, and generate recommended response actions. The algorithms operating within the system are responsible not only for reading log data, but also for interpreting incidents, evaluating threat levels, and associating related records within a centralized database.

The analysis mechanism implemented in the system is structured under three main categories by considering data collection, AI-assisted incident classification, and threat assessment processes.[9]

A. Collection of SIEM Logs and Structural Data Analysis

In the first analysis layer of the system, incident information entered by users through Google Forms is collected in a standardized structure. Through the form, users submit their full name, email address, department information, incident title, priority level, and SIEM log content.

Once the form is submitted, the data is automatically transferred to the Google Sheets table. This structure reduces irregular data entry and ensures that all records are collected through predefined fields.

The Google Sheets Trigger node detects newly added rows and starts the workflow process automatically. Then, the incoming data is checked by n8n and the required field mapping is completed before transferring the records to the Airtable database.

The main objective of this stage is to collect security incidents in an organized manner and create a reliable data source for the following analysis steps. As a result, the data flow is standardized and the processing sequence proceeds automatically.[10]

B. AI-Assisted Log Classification and Risk Score Analysis

In the core analysis layer of the system, SIEM log records are sent to the Groq-based artificial intelligence model through the AI Agent node in n8n.

The AI model analyzes the log content using a natural language processing approach and interprets the incident from a cybersecurity perspective. The generated output includes an incident summary, attack type classification, risk score, and recommended response actions.

During the risk evaluation process, indicators such as failed login attempts, suspicious connections, access violations, unusual user activities, and malware-related behavior are considered.

For example, records containing a large number of failed login attempts within a short period are classified as brute force attacks and assigned a high-risk score. Similarly, suspicious file execution activities are classified as malware activity, unusual logins from different countries are classified as suspicious login, and intensive traffic patterns are evaluated as DDoS attacks.

The results generated by the analysis are then processed through the Edit Fields node and transferred to the database as structured fields.[11]

C. Threat Intelligence and Centralized Threat Assessment

The analysis results generated by the artificial intelligence model are not only stored within the incident record. The system also stores historical records centrally through the Threat Intelligence and AI Analysis History tables.

The Threat Intelligence table stores attack type, threat description, and recommended security actions. This allows security teams to review previously detected incidents and compare similar threats more effectively.[12]

The AI Analysis History table stores the outputs generated by the artificial intelligence model. In this way, each log record, generated analysis, and assigned risk score are preserved for future reference.

Linked record fields in Airtable are used to associate incident records with user information and AI analysis history. This makes it possible to view all related information for a single security incident from a centralized structure.

Through this approach, the system goes beyond real-time incident analysis and becomes an integrated security automation platform that stores historical records, correlates threats, and centralizes incident management.[13]

IV. FINDINGS AND EVALUATION

In order to evaluate the accuracy and operational performance of the developed n8n-based AI-assisted SIEM log analysis system, several test scenarios involving different cybersecurity incidents were conducted. During the testing process, multiple SIEM log records representing different attack types were entered into the system and the outputs generated by the artificial intelligence model were examined.

The test results demonstrated that the system was able to correctly identify attack types, generate appropriate risk scores, and automatically send notifications to users.[14]

A. Application and Test Scenario

During the testing phase, SIEM log records representing various security incidents such as brute force attacks, malware activity, suspicious VPN connections, unauthorized access attempts, and DDoS traffic were prepared and submitted to the system through Google Forms.

The submitted records were automatically transferred to the Google Sheets table. Afterward, the Google Sheets Trigger node

in n8n detected the new row and automatically initiated the workflow process.

Once the workflow was triggered, the related SIEM log was forwarded to the Groq artificial intelligence model through the AI Agent node. The AI model analyzed the log content, identified the attack type, generated an incident summary, and assigned a risk score.

The conducted tests showed that the system successfully filtered unnecessary details in the log records and correctly focused on security-relevant information. Events that would normally require a long manual review process were analyzed by the system within a short period and meaningful outputs were generated.

As a result of these tests, it was observed that the developed system could process security incidents quickly and accurately.

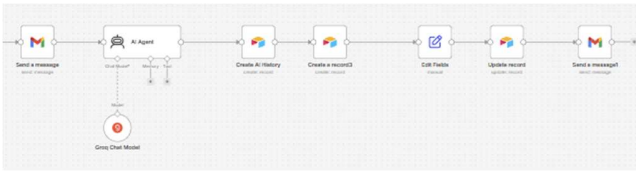


Fig. 2. SIEM log records processed through the n8n workflow and AI Agent analysis process

B. Database Layer and Notification Process

After the AI analysis process was completed, the storage of generated results in the database and user notification stages were tested.

Through Airtable integration, incident title, user information, attack type, risk score, AI-generated summary, and recommended response actions were automatically recorded in the related tables.

While the Incidents table stored incident records, the AI Analysis History table preserved the analysis history and the Threat Intelligence table centrally stored threat-related information.

Additionally, the Gmail node integrated into the workflow was successfully tested. When an incident was submitted to the system, the first automated email informing the user that the request had been received was sent. After the analysis was completed, a second email containing the incident summary and recommended response actions was delivered.

These results demonstrate that the system is not only an analysis platform, but also an integrated security automation solution that provides record management and automated user notifications.

	Analysis ID	Incident	SI Prompt Used	SI Groq Response	SI Analysis Time
1	12	96	Mesur saatleri dnynda cok s...	Q021: Mesur saatleri dnynd...	17.3.2025 11:52pm
2	14	98	Iç agdakti sistemler ararında...	Q021: İç agdakti sistemler ar...	18.3.2025 12:58am

Fig. 3. Analysis results stored in the Airtable database and automatic email notifications sent to users

C. Discussion and System Limitations

Although the developed system produced successful results in analyzing security incidents and automating incident management, several technical limitations remain due to the current architecture.

Processing load under high-volume logs: During periods of intensive network traffic or active attacks, a large number of security incidents may occur within a short time. In such cases, processing each record sequentially through the n8n workflow may create additional workload on the system. In future versions, performance may be improved through queue management or parallel processing structures.

Variations in AI-based interpretation: Since SIEM logs may appear in different formats and structures, the interpretation generated by the AI model may vary in certain cases. Therefore, more advanced prompt structures or support for multiple AI models may improve analysis accuracy.

Dependency on external services: The combined use of services such as Google Forms, Airtable, and Gmail provides flexibility and convenience; however, connection issues or API-based limitations may occur. For this reason, a more independent infrastructure may be developed in future studies.

Lack of direct SIEM platform integration: The current system was tested using manually entered log records. Direct integration with real-world security platforms such as Splunk, Wazuh, or Elastic SIEM could expand the practical applicability of the system.

Overall evaluation showed that the developed system can centrally manage security incidents, perform rapid analysis with AI assistance, and successfully automate user notification processes.

V. CONCLUSION

Manual examination of SIEM log records and evaluation of security incidents is a time-consuming and attention-intensive process. Especially in systems where a large number of logs are generated, analyzing incidents one by one increases operational workload and may delay the detection of critical security threats. In this study, an n8n-based AI-assisted security automation system was developed to automate this process and enable faster evaluation of security incidents.

In the developed system, SIEM log records submitted by users through Google Forms were collected through Google Sheets and automatically processed via the n8n workflow platform. Using the AI Agent node within the workflow and a Groq-based artificial intelligence model, log records were analyzed automatically and incident summaries, attack types, risk scores, and recommended response actions were generated.

During the testing phase, sample log records representing different security incidents such as brute force attacks, malware activity, suspicious login, unauthorized access attempts, and DDoS traffic were submitted to the system. The test results showed that the system could analyze log records within a short period, correctly identify the incident type, and generate appropriate risk evaluations.

In addition, the system was not limited to analysis only; it also automatically stored the generated results in the Airtable database and sent notification emails to users through Gmail integration. This enabled the incident management process to be monitored centrally while keeping users informed throughout every stage of the workflow.

Previous studies indicate that AI-based security systems can significantly reduce manual intervention, although they may not completely eliminate it. Similarly, the structure developed in this study provides a practical automation solution that supports security analysts in decision-making processes and reduces incident evaluation time.

In summary, this project demonstrates that SIEM log analysis processes can be performed faster and in a more organized manner by combining artificial intelligence, workflow automation, and a centralized database structure. The developed system accelerated security operations, reduced manual workload, and improved the efficiency of incident management.

REFERENCES

- [1] G. González-Granadillo, S. González-Zarzosa, and R. Diaz, "Security information and event management (SIEM): Analysis, trends, and usage in critical infrastructures," *Sensors*, vol. 21, no. 14, Jul. 2021, doi: 10.3390/s21144759.
- [2] M. Sabri Elmastaş, "Journal of Aeronautics and Space Technologies 16(2) (2023) 36-55 Detection of Current Attacks in Active Directory Environment with Log Correlation Methods Aktif Dizin Ortamındaki Güncel Saldırıların Log Korelasyon Yöntemleri ile Tespiti," 2023.
- [3] A. G. Sadowski, K. Kavanagh, and T. Bussa, "Licensed for Distribution Critical Capabilities for Security Information and Event Management," 2020. [Online]. Available: <https://www.gartner.com/doc/reprints?id=1-1YHB12KL&ct=200226&st=sb>
- [4] R. O. Kryukov, E. V. Fedorchenko, I. V. Kotenko, E. S. Novikova, and V. M. Zima, "Security assessment based on attack graphs using NVD and MITRE ATT & CK database for heterogeneous infrastructures," *Informatsionno-Upravliaiushchie Sistemy*, no. 2, pp. 39–50, 2024, doi: 10.31799/1684-8853-2024-2-39-50.
- [5] A. Akpınar, "VERİ MERKEZİ KATMANLI GÜVENLİK TASARIMI İLE ETKİN OLAY ANALİZİ VE YÖNETİMİ YÜKSEK LİSANS TEZİ," 2020.
- [6] T. Oğuz, "TÜRKİYE CUMHURİYETİ ANKARA ÜNİVERSİTESİ SOSYAL BİLİMLER ENSTİTÜSÜ BİLGİ VE BELGE YÖNETİMİ ANABİLİM DALI KÜTÜPHANE ARAMA MOTORLARINDA LOG ANALİZLERİ İLE KULLANICI TANIMA VE DERME GELİŞTİRME YÜKSEK LİSANS TEZİ İSMAİL DURDU TEZ DANIŞMANI," 2019.
- [7] P. Danquah, "Security Operations Center: A Framework for Automated Triage, Containment and Escalation," *Journal of Information Security*, vol. 11, no. 04, pp. 225–240, 2020, doi: 10.4236/jis.2020.114015.
- [8] F. Pavanello, M. Giaimo, W. Phoenix, and S. Cagol, "OSINT-based Email Analyzer for Phishing Detection," 2023.
- [9] F. C. J. I. Moenen *et al.*, "Screening for platelet function disorders with Multiplate and platelet function analyzer," *Platelets*, vol. 30, no. 1, pp. 81–87, Jan. 2019, doi: 10.1080/09537104.2017.1371290.
- [10] E. Gül +1 and N. Yılmaz, "LOG MANAGEMENT WITH OPEN SOURCE TOOLS," 2019.
- [11] C. V. Garware and S. N. Zisad, "OpenSOC-AI: Democratizing Security Operations with Parameter Efficient LLM Log Analysis," Apr. 2026, [Online]. Available: <http://arxiv.org/abs/2604.26217>
- [12] A. Rot and B. Olszewski, "Advanced Persistent Threats Attacks in Cyberspace. Threats, Vulnerabilities, Methods of Protection," in *Position Papers of the 2017 Federated Conference on Computer Science and Information Systems*, PTI, Sep. 2017, pp. 113–117. doi: 10.15439/2017f488.
- [13] M. R. Fatemi, A. A. Ghorbani, and J. J. McNally, "Threat-Hunting in Windows Environment Using Host-based Log Data," 2017.
- [14] E. De Posgrados "espog, F. E. Vela Vela, M. Pablo, M. Recalde, and V. Quito -Ecuador, "UNIVERSIDAD TECNOLÓGICA ISRAEL Título del artículo ESTUDIO DE UN SECURITY INFORMATION AND EVENT MANAGEMENT (SIEM) PARA BASE DE DATOS SQL SERVER CASO DE ESTUDIO: MINISTERIO DE RELACIONES EXTERIORES Y MOVILIDAD HUMANA Línea de Investigación: SEGURIDAD INFORMÁTICA Campo amplio de conocimiento: TECNOLOGÍAS DE LA INFORMACIÓN Y COMUNICACIÓN Autor," 2022.